Lecture 23

Yang–Mills theory and instantons

Objectives. *Lecture 21 supplied the geometry of a gauge field. We now choose an action and derive its field equation. For an* SU(2) *connection extending over the four-sphere, the curvature also gives an integer. This integer bounds the action and identifies solutions of a first-order equation. The BPST field fixes the orientation and normalization in a concrete example.*

By the end of this lecture you will be able to:

- Derive the vacuum Yang–Mills equation with the fundamental trace normalization.
- Prove the local Chern–Simons identity and distinguish it from global exactness.
- Complete the Euclidean action square and identify its equality condition.
- Relate BPST duality and Chern number to the chosen orientation.

23.1 A connection on the principal bundle

Lecture 20's principal bundle $P \to M$ carries a right action of a Lie group G. Its Lie algebra is $\mathfrak{g}$. For $\xi \in \mathfrak{g}$, the *fundamental vertical vector* at $p \in P$ is $\xi_p^{\#} = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_0 p \exp(t\xi)$. It is tangent to the fibre.

Remark 23.1 (Principal connections)**.** A *principal connection* is a smooth $\mathfrak{g}$-valued one-form ω_P on P with

$$\omega_P(\xi^{\#}) = \xi, \qquad R_a^* \omega_P = \mathrm{Ad}_{a^{-1}}\, \omega_P \quad (a \in G),$$

where $R_a(p) = pa$ and $\mathrm{Ad}_a\, \xi = a\xi a^{-1}$ for a matrix group. Its kernel gives the horizontal tangent spaces, complementary to the vertical tangent spaces. In a local section $s_i \colon U_i \to P$, the potential is $A_i = s_i^* \omega_P$. For $s_j = s_i g_{ij}$ these potentials obey Lecture 21's frame law. A representation $\rho \colon G \to \mathrm{GL}(V)$ induces on the associated bundle the local derivative $\mathrm{d} + \rho_*(A_i)$, where ρ_* is the representation's differential at the identity. Thus one principal connection supplies the potentials in all representations. These intrinsic-to-local facts are cited from Tu, §28.1, §29.4 and §31.3–31.5.

Gauge transformations are bundle automorphisms over the identity that commute with the right action. Lecture 21 represents them by compatible maps $u_j = g_{ij}^{-1} u_i g_{ij}$. A global map $M \to G$ suffices in a chosen trivialization. The passive frame change $e' = eh$ has $A^h = h^{-1}Ah + h^{-1}\mathrm{d}h$. All calculations below use this convention.

23.2 The Yang–Mills action and equation

Use a connection for SU(r), with $r \geq 2$, in its fundamental representation. Choose anti-Hermitian generators T^a normalized by $\mathrm{tr}(T^a T^b) = -\delta^{ab}/2$. For SU(2), these are $T^a = -\mathrm{i}\sigma_a/2$

with the Pauli matrices of Lecture 21. The real basis from *Lie Groups and Their Representation Theory* is related to these generators by $(\mathrm{i}\sigma_z, \mathrm{i}\sigma_y, \mathrm{i}\sigma_x) = (-2T^3, -2T^2, -2T^1)$. That basis has cyclic bracket factor 2. The generators T^a used here have cyclic bracket factor one. In Minkowski coordinates $x^0 = t$, the metric has signature $(+---)$ and the volume orientation is $\mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. The real component coefficients are defined by

$$A = T^a A^a_\mu \mathrm{d}x^\mu, \qquad F = \mathrm{d}A + A \wedge A = \frac{1}{2} T^a F^a_{\mu\nu} \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.$$

Here a runs over the $r^2 - 1$ generators. Spacetime indices run from 0 to 3. The coupling $g_{\mathrm{YM}} > 0$ is a scalar. Assume the displayed action integrals converge, or work on a bounded region with variations supported in its interior. The *Yang–Mills action* is the following metric-dependent functional.

Theorem 23.2 (The field equation · *proved*)**.** *The Minkowski action*

$$S_M[A] = \frac{1}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(F \wedge \star F) = -\frac{1}{4 g_{\mathrm{YM}}^2} \int_M F^a_{\mu\nu} F^{a\mu\nu} \operatorname{vol}$$

is gauge invariant. Its stationary points under arbitrary smooth compactly supported connection variations satisfy $D\star F = 0$. For an endomorphism-valued k-form B, the derivative here is $DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A$.

Proof. The Hodge pairing of two two-forms contracts their components with a factor $1/2$. Multiplying this by $\operatorname{tr}(T^a T^b) = -\delta^{ab}/2$ gives the coefficient $-1/4$. On overlaps and under a frame change, $F^h = h^{-1} F h$. The star acts only on differential forms, so conjugation and cyclicity of the trace make the integrand global and gauge invariant.

The difference of two connections transforms by conjugation, with no derivative term. Such Lie-algebra-valued forms are called *adjoint-valued*. Let $a = \delta A$ be a compactly supported adjoint-valued one-form. Differentiate $\mathrm{d}(A + \varepsilon a) + (A + \varepsilon a)^2$ at $\varepsilon = 0$. The result is $\delta F = \mathrm{d}a + A \wedge a + a \wedge A = Da$. Symmetry of the two-form pairing gives

$$\delta S_M = \frac{2}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(Da \wedge \star F),$$
$$\operatorname{tr}(Da \wedge \star F) = \mathrm{d} \operatorname{tr}(a \wedge \star F) + \operatorname{tr}(a \wedge D\star F).$$

The second line follows from the graded product rule. The connection commutators cancel inside the trace. Stokes removes the compactly supported exact term. The remaining pairing with every a vanishes exactly when $D\star F = 0$. The trace pairing is nondegenerate on $\mathfrak{su}(r)$, so it detects every component. ∎

With matter, define the connection-valued current three-form $\mathcal{J}_{\mathrm{conn}}$ by $\delta S_{\mathrm{matter}} = -(2/g_{\mathrm{YM}}^2) \int_M \operatorname{tr}(a \wedge \mathcal{J}_{\mathrm{conn}})$. The same variation then gives $D\star F = \mathcal{J}_{\mathrm{conn}}$.

For U(1) the generator is $-\mathrm{i}$, whose trace square is -1. Use the different prefactor $1/(2g^2)$, where $g = q/\hbar \neq 0$ is signed. Lecture 21's physical ledger gives

$$A = -\frac{\mathrm{i}q}{\hbar} A_{\mathrm{em}}, \qquad F = -\frac{\mathrm{i}q}{\hbar} F_{\mathrm{em}}, \qquad S_M = \frac{1}{2g^2} \int_M F \wedge \star F = -\frac{1}{2} \int_M F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}.$$

Let $\mathcal{J} = \star j$ be Lecture 15's real current and set $\mathcal{J}_{\mathrm{conn}} = -(\mathrm{i}q/\hbar)\mathcal{J}$. For this prefactor, with $a = \delta A$, define the matter current by

$$\delta S_{\mathrm{matter}} = -\frac{1}{g^2} \int_M a \wedge \mathcal{J}_{\mathrm{conn}} = \int_M \delta A_{\mathrm{em}} \wedge \mathcal{J}.$$

The equation $\mathrm{d}{\star}F = \mathcal{J}_{\mathrm{conn}}$ becomes Lecture 15's $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. The real current and the imaginary connection current are distinct.

The Hermitian-potential dictionary is $A = -\mathrm{i}A_{\mathrm{Tong}}$ and $F = -\mathrm{i}F_{\mathrm{Tong}}$ for Tong's unrescaled matrices. His rescaling $A_{\mathrm{phys}} = A_{\mathrm{Tong}}/g_{\mathrm{YM}}$ instead gives $A = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$. Tong, §2.1, uses the positive Hermitian trace in his component action.

23.3 Euclidean action and duality

Replace the Minkowski metric by a positive Riemannian metric on an oriented four-manifold. The *Euclidean action* is

$$S_E[A] = -\frac{1}{g_{\mathrm{YM}}^2}\int_M \operatorname{tr}(F \wedge {\star}F).$$

The minus sign makes the density nonnegative. In an orthonormal form basis, the matrix pairing is $-\operatorname{tr}(X^2) \geq 0$ for anti-Hermitian X. We use Euclidean time $\tau = x^4$ and spatial-first orientation $\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3 \wedge \mathrm{d}x^4$ on $\mathbb{R}^4$. It is the negative of the time-first orientation.

Lecture 15 gives ${\star}^2 = 1$ on Euclidean two-forms. A curvature is *self-dual* if $F = {\star}F$ and *anti-self-dual* if $F = -{\star}F$. Its orthogonal parts $F_+ = (F + {\star}F)/2$ and $F_- = (F - {\star}F)/2$ obey ${\star}F_+ = F_+$ and ${\star}F_- = -F_-$. Either duality equation implies the vacuum field equation, since $D{\star}F = \pm DF = 0$ by Lecture 21's Bianchi identity. Real Minkowski two-forms have ${\star}^2 = -1$. Imposing either real duality equation there forces $F = 0$.

Reversing orientation changes the star's sign and also changes integration's sign. Their product leaves S_E unchanged. The action depends on the metric through the star. An integral of a fixed four-form instead changes sign with orientation.

23.4 Chern–Simons and the second Chern number

For matrix-valued forms of degrees p and q, cyclicity becomes $\operatorname{tr}(B \wedge C) = (-1)^{pq}\operatorname{tr}(C \wedge B)$. This follows by interchanging the scalar differential forms in each matrix entry. In a local frame define the *Chern–Simons three-form* by

$$\mathrm{CS}(A) = \operatorname{tr}\left(A \wedge \mathrm{d}A + \frac{2}{3}A \wedge A \wedge A\right).$$

Theorem 23.3 (The local transgression · *sketched; local identities proved*)**.** *The local forms satisfy*

$$\mathrm{d}\,\mathrm{CS}(A) = \operatorname{tr}(F \wedge F), \qquad \mathrm{CS}(\vartheta_{\mathrm{MC}}) = -\frac{1}{3}\operatorname{tr}(\vartheta_{\mathrm{MC}}^3), \qquad \vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h.$$

For a general frame change, the local gauge law is

$$\mathrm{CS}(A^h) = \mathrm{CS}(A) - \mathrm{d}\operatorname{tr}(\mathrm{d}hh^{-1} \wedge A) - \frac{1}{3}\operatorname{tr}(\vartheta_{\mathrm{MC}}^3).$$

The resulting global characteristic-class interpretation is cited.

Proof of the local identities. Write matrix wedge products as juxtaposition. The product rule gives $\mathrm{d}\operatorname{tr}(A\mathrm{d}A) = \operatorname{tr}((\mathrm{d}A)^2)$. For the cubic term it gives

$$\mathrm{d}\operatorname{tr}(A^3) = \operatorname{tr}(\mathrm{d}A\,A^2 - A\,\mathrm{d}A\,A + A^2\mathrm{d}A) = 3\operatorname{tr}(\mathrm{d}A\,A^2).$$

Moving the first A past the remaining total degree three changes sign. This proves the middle term's sign and also gives $\operatorname{tr}(A^4) = -\operatorname{tr}(A^4) = 0$. Consequently

$$\operatorname{tr}(F^2) = \operatorname{tr}((\mathrm{d}A)^2 + 2\mathrm{d}A\,A^2) = \mathrm{d}\,\mathrm{CS}(A).$$

The Maurer–Cartan equation is $\mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}}^2$. Substitution into the definition gives $(-1 + 2/3)\operatorname{tr}(\vartheta_{\mathrm{MC}}^3)$, which is the claimed pure-gauge value. ∎

For the general law set $B = h^{-1}Ah$. Expand $\mathrm{CS}(B+\vartheta_{\mathrm{MC}})$ using $\mathrm{d}B = h^{-1}\mathrm{d}Ah - \vartheta_{\mathrm{MC}}B - B\vartheta_{\mathrm{MC}}$. Graded trace cyclicity groups the mixed terms into $-\mathrm{d}\operatorname{tr}(\vartheta_{\mathrm{MC}} \wedge B) = -\mathrm{d}\operatorname{tr}(\mathrm{d}hh^{-1} \wedge A)$. The unmixed terms give the two Chern–Simons forms above. This outlines the expansion. Nakahara, §11.5, supplies the general transgression framework. The local potential and its Chern–Simons form need not be global. In particular, the local identity does not make every closed-manifold Chern number zero.

For an SU(2) bundle over an oriented closed four-manifold, Lecture 22 gives its *second Chern number*

$$k = \int_M c_2(F) = \frac{1}{8\pi^2}\int_M \operatorname{tr}(F \wedge F).$$

It is independent of the connection and metric. Its sign uses the orientation. The *Chern character* is the trace exponential $\mathrm{ch}(F) = \operatorname{tr}\exp(\mathrm{i}F/(2\pi))$, expanded in wedge powers. The quadratic exponential term gives its degree-four component on an SU(2) bundle:

$$\mathrm{ch}_2(F) = -\frac{1}{8\pi^2}\operatorname{tr}(F \wedge F) = -c_2(F).$$

Thus our k is minus the integrated second Chern character. Nakahara's (11.54)–(11.55) and (12.90) use the latter as their instanton number.

On S^4 the integer value of our Chern number can be read from an equatorial transition. Represent the two hemispheres by an inside and an outside four-ball as in the figure.

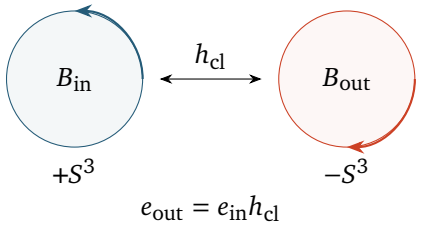


Figure 23.1: Two local Stokes integrals have opposite equatorial orientations. The frame order is $e_{\mathrm{out}} = e_{\mathrm{in}}h_{\mathrm{cl}}$. The equator is oriented as ∂B_{in}.

Theorem 23.4 (The clutching integer · *sketched; global construction cited*). *Let* $h_{\mathrm{cl}} \colon S^3 \to \mathrm{SU}(2)$ *glue the frames in the figure's order. Orient* SU(2) *as the outward unit sphere under* $(a_1, a_2, a_3, a_4) \mapsto a_4 I + \mathrm{i}a_a\sigma_a$. *Then* $k = -\deg h_{\mathrm{cl}} \in \mathbb{Z}$.

Sketch. Use local frames over the two balls, and Stokes with their opposite boundaries. The gauge law and the vanishing integral of an equatorial exact form give

$$k = \frac{1}{8\pi^2}\oint_{S^3}\bigl(\mathrm{CS}(A_{\mathrm{in}}) - \mathrm{CS}(A_{\mathrm{out}})\bigr) = \frac{1}{24\pi^2}\oint_{S^3}\operatorname{tr}\bigl((h_{\mathrm{cl}}^{-1}\mathrm{d}h_{\mathrm{cl}})^3\bigr).$$

For the identity unit-quaternion map, at $a_4 = 1$ the Maurer–Cartan form is $\mathrm{i}\sigma_a\mathrm{d}a_a$. The Pauli products give cubic trace $12\mathrm{d}a_1 \wedge \mathrm{d}a_2 \wedge \mathrm{d}a_3$. The outward sphere volume there is its negative,

since the outward normal is last in the ordered ambient coordinates. Both forms are left invariant, so everywhere $\operatorname{tr}(\vartheta_{\mathrm{MC}}^3) = -12\,\mathrm{vol}_{S^3}$. The sphere volume is $4\pi\int_0^\pi \sin^2\chi\,\mathrm{d}\chi = 2\pi^2$. Thus the normalized three-form $-\operatorname{tr}(\vartheta_{\mathrm{MC}}^3)/(24\pi^2)$ has integral one. Lecture 14's degree formula identifies the displayed integral as minus the degree. The global two-ball construction is the cited input from Nakahara, §10.5.5. ∎

On a region with boundary, the integral of the Chern form need not be an integer, just as the cap flux in Lecture 22 need not be integral.

23.5 The action bound and BPST

For an adjoint-valued two-form B, write $\|B\|^2 = -\int_M \operatorname{tr}(B\wedge\star B)$. This is nonnegative and vanishes only for $B = 0$.

Theorem 23.5 (The instanton bound · *proved*). *For an* SU(2) *connection over Euclidean* S^4,

$$S_E \geq \frac{8\pi^2}{g_{\mathrm{YM}}^2}|k|.$$

Equality holds exactly for a self-dual or anti-self-dual curvature. In our negative-trace convention, a nonzero self-dual field has $k < 0$, *and a nonzero anti-self-dual field has* $k > 0$.

Proof. The Euclidean star is an isometry with square one on two-forms. Expand the square for either $s = +1$ or $s = -1$ to obtain

$$-\operatorname{tr}(F\wedge\star F) = -\frac{1}{2}\operatorname{tr}\big((F - s\star F)\wedge\star(F - s\star F)\big) - s\operatorname{tr}(F\wedge F).$$

After integration this gives $g_{\mathrm{YM}}^2 S_E = \frac{1}{2}\|F - s\star F\|^2 - s8\pi^2 k$. Choose s so that $-sk = |k|$. Nonnegativity proves the bound. Equality requires $F = s\star F$. Conversely, that equation removes the square and gives equality. For $k = 0$ equality forces $F = 0$, which obeys both duality equations. For a nonzero dual field, $8\pi^2 k = -s\|F\|^2$, proving the stated signs. ∎

An *instanton* here is a smooth finite-action Euclidean connection with self-dual or anti-self-dual curvature. On S^4, or under the stated extension from $\mathbb{R}^4$, the bound makes it an action minimum within its fixed integer sector. The *BPST instanton* is the following charge-one SU(2) example.

Proposition 23.6 (BPST normalization · *sketched*). *For size* $\lambda > 0$ *on spatial-first* $\mathbb{R}^4$, *the regular BPST field is anti-self-dual, extends over the conformal compactification* S^4, *and has* $k = +1$ *and* $S_E = 8\pi^2/g_{\mathrm{YM}}^2$. *Reflection of one coordinate gives its self-dual partner with* $k = -1$ *and the same action.*

The extension over the compactification is a cited part of the BPST construction in Nakahara, §10.5.5, and Tong, §2.3.1. We assume this asymptotic regularity when treating an $\mathbb{R}^4$ field as a connection on S^4. Finite action alone is not used as a proof of that extension. The Euclidean star on two-forms is unchanged by a conformal metric rescaling, because its scale factor in dimension four is the zeroth power. Thus stereographic compactification preserves the duality equation and action. Give $S^4 \subset \mathbb{R}^5$ its outward orientation. Let $e_1,\dots,e_5$ be the standard basis of $\mathbb{R}^5$ and put $r^2 = \sum_{\mu=1}^4 (x^\mu)^2$. Use the inverse south-pole stereographic chart $X\colon \mathbb{R}^4 \to S^4 \setminus \{-e_5\}$, given by

$$X(x) = \frac{(2x^1, 2x^2, 2x^3, 2x^4, 1 - r^2)}{1 + r^2}.$$

At $x = 0$, the outward normal is e_5 and the four coordinate tangent vectors are $2e_i$. The ordered determinant is +16. The inverse $x^i = X^i/(1 + X^5)$, for $i \in \{1, 2, 3, 4\}$, shows that this is a smooth chart. Since $\mathbb{R}^4$ is connected, it preserves the spatial-first orientation everywhere. Consequently the computed charge on $\mathbb{R}^4$ is also the charge on this outward S^4.

The normalized radial contribution for this example is plotted in the figure.

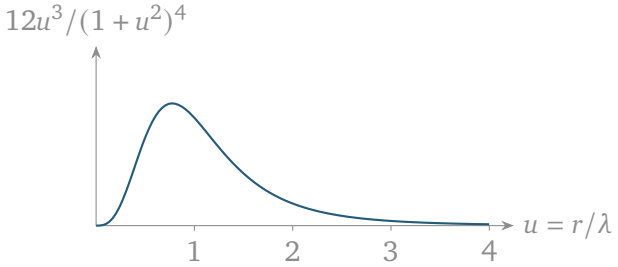


Figure 23.2: For $u = r/\lambda$, the BPST Chern-number contribution per du is $12u^3/(1 + u^2)^4$. Its integral over $[0, \infty)$ is 1. The plot shows the initial interval $0 \le u \le 4$.

23.6 A charge-one field and a topological term

Example 23.7 (The BPST action)**.** Compute the charge and action of the regular field with $\lambda > 0$. Recall $r^2 = \sum_{\mu=1}^{4} (x^\mu)^2$, with spatial indices $a, b, c \in \{1, 2, 3\}$. The potential is smooth at the origin and is given by

$$A = \frac{\mathrm{i}\sigma_a \left(x^4 \mathrm{d}x^a - x^a \mathrm{d}x^4 + \epsilon_{abc} x^b \mathrm{d}x^c\right)}{r^2 + \lambda^2}.$$

Differentiation and the Pauli commutator give its curvature

$$F = \frac{2\mathrm{i}\lambda^2 \sigma_a}{(r^2 + \lambda^2)^2} \left(\frac{1}{2} \epsilon_{abc} \mathrm{d}x^b \wedge \mathrm{d}x^c - \mathrm{d}x^a \wedge \mathrm{d}x^4 \right).$$

For example, the coefficient pair multiplying σ_1 is $\mathrm{d}x^2 \wedge \mathrm{d}x^3 - \mathrm{d}x^1 \wedge \mathrm{d}x^4$. The spatial-first star interchanges these two basis forms, so their difference is anti-self-dual. The other cyclic pairs have the same property. Taking the fundamental trace gives

$$\operatorname{tr}(F \wedge F) = -\operatorname{tr}(F \wedge \star F) = \frac{48\lambda^4}{(r^2 + \lambda^2)^4} \operatorname{vol}_{\mathbb{R}^4}.$$

Radial integration, followed by $w = r^2/\lambda^2$, yields

$$\int_{\mathbb{R}^4} \operatorname{tr}(F \wedge F) = 96\pi^2 \lambda^4 \int_0^\infty \frac{r^3 \mathrm{d}r}{(r^2 + \lambda^2)^4} = 48\pi^2 \int_0^\infty \frac{w \mathrm{d}w}{(1 + w)^4} = 8\pi^2.$$

The last integral is $1/6$, by writing $w = (1 + w) - 1$ and integrating the two powers. Consequently $k = 1$ and $S_E = 8\pi^2/g_{\mathrm{YM}}^2$, independent of the size.

For $r > 0$, define the asymptotic unit-quaternion map $h_\infty = (x^4 I + \mathrm{i}x^a \sigma_a)/r$. The same potential is $A = [r^2/(r^2 + \lambda^2)] h_\infty^{-1} \mathrm{d}h_\infty$. The map has degree +1 in the stated orientation. At infinity the pure-gauge Chern–Simons integral therefore gives $k = -\int_{S^3} \operatorname{tr}((h_\infty^{-1}\mathrm{d}h_\infty)^3)/(24\pi^2) = +1$. The clutching map has the inverse order. The change of frame $h_{\mathrm{cl}} = h_\infty^{-1}$ makes the outer asymptotic potential zero. Its degree is -1, so the clutching formula gives the same charge. Lecture 4 wrote $U = aI - \mathrm{i}v^a \sigma_a$. Our oriented quaternion coordinates there are $(a_4, \mathbf{a}) = (a, -\mathbf{v})$. Reversing to time-first orientation makes this same field self-dual and changes k to -1.

For a fixed real angle θ_{YM}, the *theta term* is $S_\theta = \theta_{\mathrm{YM}} \int_M \operatorname{tr}(F \wedge F)/(8\pi^2) = \theta_{\mathrm{YM}} k$. Locally its density is $\theta_{\mathrm{YM}} \mathrm{d}\,\mathrm{CS}(A)/(8\pi^2)$. Its variation is $\delta \operatorname{tr}(F \wedge F) = 2\mathrm{d} \operatorname{tr}(a \wedge F)$, using $DF = 0$. It therefore leaves the field equation unchanged for compactly supported variations, or on a closed manifold. On a manifold with boundary, boundary conditions still matter.

Self-check. Show directly that $\operatorname{tr}(F \wedge \star F)$ is invariant under a frame change.

Answer. The transformed product is $h^{-1}(F \wedge \star F)h$, since the star commutes with matrix conjugation. Its trace equals the original trace by cyclicity.

Summary. The fundamental anti-Hermitian trace fixes the Minkowski prefactor and gives $D\star F = 0$. For U(1) the different trace normalization recovers Maxwell's action. Locally $\mathrm{d}\,\mathrm{CS}(A) = \operatorname{tr}(F \wedge F)$. The Chern–Simons form need not be global. On S^4, $k = -\deg h_{\mathrm{cl}}$ in the stated two-frame order. The Euclidean bound is $S_E \geq 8\pi^2|k|/g_{\mathrm{YM}}^2$, saturated exactly by dual fields. Regular spatial-first BPST is anti-self-dual with $k = +1$. The action needs the metric and is orientation independent. The Chern number has the opposite dependence. Lecture 24 returns to line-bundle connections generated by quantum eigenstates.

Reading. Tu, *Differential Geometry*, §28.1, §29.4 and §31.3–31.5. Nakahara, §10.5.4–10.5.5, §11.3.1 (11.54)–(11.55), §11.5 and §12.6 (12.90). Tong, *Gauge Theory*, §2.1 and §2.3.1, pp. 28–30 and 52–55. In Nakahara (10.124)–(10.125), the equatorial north-pole volume is $+\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. Our outward volume form at that pole is its negative. Tong calls his time-first BPST field self-dual with $\nu = +1$. Our spatial-first orientation and translated anti-Hermitian field give anti-self-duality and $k = +1$. Use the orientation, potential and Chern-character dictionaries together.